

Advise the patient with diabetes

- Help the patient to manage his/her CVD risk ↗ 129. Educate on foot care to prevent ulcers and amputation ↗ 66.
- Discuss diet: avoid white/brown sugar and honey, use artificial sweetener instead. Cut down on starch (rice, noodles, bread, potato, sweet potato, butternut, mielies, pap, samp).
- Explain importance of adherence and to eat regular meals. If difficulty with adherence, give adherence support ↗ 173 and refer to a community health worker, if available.
- Ensure patient can recognise and manage hypoglycaemia (shaking, sweating, palpitations, weakness, hunger):
 - Drink milk with sugar or eat a sweet. Always carry something sweet. If not in clinic and fits, confusion or coma, rub sugar inside mouth and call ambulance. Go to clinic if illness (like diarrhoea).
 - Identify and manage the cause: increased exercise, missed meals, inappropriate dosing of glucose-lowering medications, alcohol, infections.
- Advise patient that s/he is at risk of severe COVID-19 disease and should adhere strictly to physical distancing, good hand/respiratory hygiene and keep up to date with his/her COVID-19 vaccinations.
- If on/starting insulin: discuss injection technique/sites (abdomen, thighs, arms), store insulin in fridge/cool dark place, meal frequency, symptoms of hypoglycaemia/hyperglycaemia, disposal of sharps. Injection technique and sites (abdomen, thighs, arms), store insulin in fridge/cool dark place, meal frequency, recognising hypoglycaemia/hyperglycaemia, sharps disposal at clinic.
- Advise that if unwell and vomiting/not eating as usual: to increase fluid intake, check glucose 3 times a day if possible and adjust insulin dose if necessary (avoid stopping insulin).

Treat the patient with diabetes

- Check that patient is up to date with COVID-19 vaccine. If age > 65 years, or known HIV or heart or lung disease, give **influenza vaccine** 0.5mL IM yearly.
- If **known CVD**¹: give **simvastatin**² 40mg³ at night and **aspirin** daily. Avoid simvastatin if pregnant and avoid aspirin if peptic ulcer, dyspepsia, kidney disease. Avoid both if liver disease.
- If **no known CVD**¹ but CVD risk > 20%, eGFR < 60, known with diabetes > 10 years or age > 40 years, give **simvastatin**² 10mg at night. Avoid if pregnant or liver disease.
- If albuminuria/proteinuria, give **enalapril**⁴ 5mg 12 hourly, regardless of BP. If proteinuria persists and systolic BP > 100, increase up to 10mg 12 hourly, if tolerated.
- Give glucose-lowering medication using stepwise approach as in table below. Ensure patient is adherent before increasing treatment.

Step	Medication	Breakfast	Supper	Bedtime (before 22h00)	Note
1	Metformin	500mg 500mg 850mg 1g	500mg 850mg 1g		• Avoid if eGFR < 30, liver disease, uncontrolled heart failure, alcoholism. • If on dolutegravir or eGFR 30-45, halve dose, up to maximum of metformin 500mg 12 hourly. • Take with meals. May cause self-limiting nausea, abdominal cramps or diarrhoea. Advise not to stop treatment. • Increase monthly if fasting glucose > 8 (or postprandial⁵ glucose > 10) or HbA_{1c} > 8%, and patient is adherent. • If up to 2g needed daily, metformin may be given as 850mg 8 hourly <i>instead of</i> 1g twice daily. • If after 3 months on maximum dose HbA_{1c} > 8%, move to step 2.
2	Add glimepiride (Preferred in elderly patients: > 65 years) or glibenclamide	1mg 2mg 3mg 4mg (up to 8mg) 2.5mg 5mg 5mg 5mg 7.5mg 10mg	 2.5mg 5mg 5mg 5mg		• Continue metformin. • Glimepiride: take glimepiride with breakfast. - Increase glimepiride by 1mg, at weekly intervals, up to 8mg daily if fasting glucose > 8 (or postprandial⁵ glucose > 10) or HbA_{1c} > 8%, and patient is adherent. • Glibenclamide: avoid glibenclamide if > 65 years. - Take glibenclamide 30 minutes before breakfast. Avoid missing meals. - Increase every 2 weeks if fasting glucose > 8 (or postprandial⁵ glucose > 10) or HbA_{1c} > 8%, and patient is adherent. • Avoid both in pregnancy, severe kidney (eGFR < 60) and liver disease, co-trimoxazole allergy. • If after 3 months on maximum dose HbA_{1c} > 8%, move to step 3.
3	Add basal insulin (intermediate or long acting)			Start at 10IU . If glucose remains raised, increase by 2-4units each week.	• Stop glimepiride/glibenclamide but continue metformin when starting insulin. Educate about insulin as above. • Advise patient to check glucose daily after a meal and on waking 3 times a week. Keep a record of readings. • If fasting glucose frequently > 8 (or postprandial⁵ glucose > 10), increase by 2-4units each week. • If > 20IU needed or if patient having episodes of hypoglycaemia, discuss/refer to doctor.
4	Substitute with biphasic insulin	0.2IU/kg 0.2IU/kg + 4IU 0.2IU/kg + 4IU 0.2IU/kg + 8IU 0.2IU/kg + 8IU 0.2IU/kg + 12IU	0.1IU/kg 0.1IU/kg 0.1IU/kg + 4IU 0.1IU/kg + 4IU 0.1IU/kg + 8IU 0.1IU/kg + 8IU etc		• Continue with metformin. Stop glimepiride/glibenclamide and basal insulin. Educate about insulin as above. • Start with 0.3units/kg/day. Patient to give two-thirds of total daily insulin dose 30 minutes before breakfast and one-third of total daily insulin dose 30 minutes before supper. • Advise patient to check glucose daily after a meal and on waking 3 times a week. Keep a record of readings. • If fasting glucose frequently > 8 (or postprandial⁵ glucose > 10), increase dose by 4 units each week. • If HbA_{1c} > 8% after 3 months, discuss with specialist.

Review the patient with diabetes 6 monthly once stable.

¹Cardiovascular disease (CVD) includes ischaemic heart disease, peripheral vascular disease and stroke/TIA. ²If HIV positive on lopinavir/ritonavir or atazanavir/ritonavir, avoid simvastatin, give instead **atorvastatin** 10mg at night. ³If on amlodipine, reduce **simvastatin** dose to 20mg at night. No dose adjustment needed for rosuvastatin, pravastatin, atorvastatin. ⁴Avoid in pregnancy, previous angioedema or renal artery stenosis. If persistent cough, refer to doctor to consider alternative. ⁵Two hours after eating.